Quiz-01 AI-2002 Artificial Intelligence Spring - 2026 Total Marks: 15

NAME: _____ Roll Number: _____ Section: _____

Q1: For each of the following activities, give a PEAS description of the task environment. (Any 2)
- Playing soccer.
- Exploring the subsurface of Arabian Sea.
- Shopping for used AI books on the Internet.
- Playing a tennis match.
- Practicing tennis against a wall.

Q2:.

a) Autonomous Delivery Drone:

Scenario: Imagine an autonomous drone designed to deliver packages in a city. The drone has sensors like GPS, cameras, and altitude sensors, and it can take actions such as moving up/down, navigating left/right, or dropping packages at specific locations.

Question:

- **What would the environment look like for this delivery drone?**
- **What are the drone's sensors and actuators in this scenario?**
- **How does the drone perceive changes in the environment, and what decisions would it make based on those perceptions?**

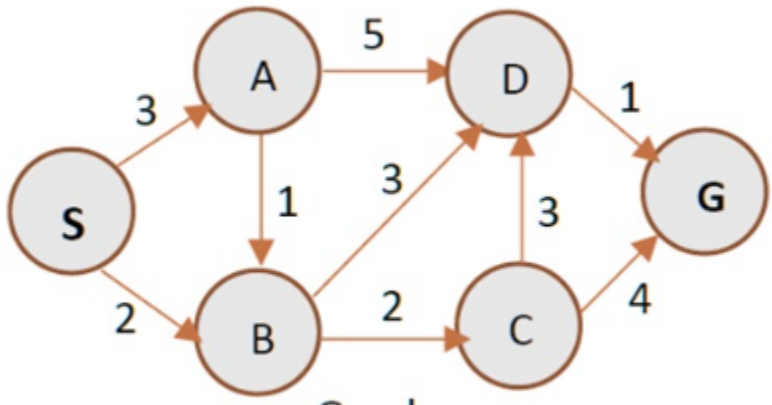
b) E-commerce Recommendation System:

Scenario: An e-commerce website uses a recommendation system to suggest products to users based on their past behavior (e.g., purchases, clicks, search history). The recommendation agent adapts to the user's preferences over time to suggest items that are most likely to be purchased.

Question:

- **What would the environment look like for this recommendation agent?**
- **How does the agent perceive changes in the environment (e.g., new user data, product preferences)?**
- **How does the agent decide which products to recommend, and what type of learning process might be involved?**

Q3: Answer the following questions about the search problem shown above. Break any ties alphabetically. For the questions that ask for a path, please show complete traversal with the help of open and closed lists.



Graph a

(a) What path would uniform cost graph search return for this search problem?
(b) What path would depth-first graph search return for this search problem?
(c) What path would A Graph search, using below given heuristic, returns for this search problem?
(d) Consider the heuristics for this problem shown in the table below.

State	h_1	h_2
S	5	4
A	3	2
B	6	6
C	2	1
D	3	3
G	0	0

i. Is h1 admissible? Yes No
ii. Is h1 consistent? Yes No
iii. Is h2 admissible? Yes No
iv. Is h2 consistent? Yes No